

PRACTICE EXERCISES



Module 1: Introduction to Databases & RDBMS Fundamentals

Module 1 - 6 Hours

Introduction to Databases & RDBMS Fundamentals

Instructions

Work on your own copy of the `icicibank` database. Paste every statement and its output into a file named `M01_<yourname>.sql` and submit it at the end of the session.

Level	Meaning	Expected time
Level 1	Recall and understanding. Answer in your own words.	25 min
Level 2	Apply. Write and run SQL against ICICI Bank data.	45 min
Level 3	Analyse and decide. Justify a design choice in writing.	30 min

Rules:

- No copying from the lab guide. If you need the syntax, look at the course material.
- Every SQL answer must include the output you actually got.
- If a statement fails, keep the error message in your answer file and explain it.

Do not delete or modify the seed data permanently. Wrap experimental changes in `BEGIN; ... ROLLBACK;` so your database is still clean for Module 2.

How much to attempt

The questions marked **(Core)** are the required set. They fit the time above, and together they cover every objective of the module. Everything unmarked is **stretch work**: attempt it if you finish early, or take it home. **You are not behind if you have completed only the core set.**

Level	Core set	Stretch
Level 1	Q1, Q2, Q4, Q5, Q7, Q8	Q3, Q6, Q9, Q10
Level 2	Q13, Q16, Q17, Q18, Q21	Q11, Q12, Q14, Q15, Q19, Q20, Q22
Level 3	Q25, Q27	Q23, Q24, Q26, Q28

Level 1 - Understanding

Q1. (Core) ICICI Bank's Baner Pune branch stores customer addresses in a spreadsheet, and the credit card department keeps its own copy of the same addresses. A customer moves house and tells only the branch. Name the two file-system problems this creates, and explain in two sentences how a database prevents both.

Q2. (Core) State three things a DBMS does that a file system does not. For each, give one concrete banking consequence of not having it.

Q3. Fill in the table with the matching term.

Formal relational term	Everyday term	ICICI Bank example
Relation	?	?
Tuple	?	?
Attribute	?	?
Degree	?	?
Cardinality	?	?

Q4. (Core) For each of the following, say whether it is a primary key, a candidate key, a foreign key, or none of these.

1. `customer.customer_id`
2. `account.account_number`
3. `account.customer_id`
4. `account.balance`
5. `employee.manager_id`
6. `branch.ifsc_code`

Q5. (Core) Classify each workload as OLTP or OLAP, and give a one-line reason.

1. A customer checks a balance on the ICICI mobile app.
2. The risk team calculates non-performing loans by branch for the quarter.
3. An ATM dispenses 10,000 rupees.
4. A regulator asks for the total value of transactions above 200,000 in the last financial year.
5. A teller opens a new savings account.
6. Marketing finds the top 100 customers by average monthly balance.

Q6. ICICI Bank's head of reporting asks to run the quarterly regulatory report directly against the core banking Oracle database, because "the data is fresher there". Give three specific technical reasons to refuse, and name what you would offer instead.

Q7. (Core) Match each ACID property to the failure it prevents.

Property	Failure
Atomicity	a. A committed transfer disappears after a power cut
Consistency	b. Two ATMs both pay out against the same balance

Property	Failure
Isolation	c. An account exists for a customer who does not
Durability	d. Money is debited but never credited

Q8. (Core) Explain in your own words why the write-ahead log is written to disk before the data page. What would break if the order were reversed?

Q9. A junior developer writes `SELECT * FROM customer` in the internet banking application. Explain which type of data independence this breaks and what will go wrong when the bank later adds a `pan_number` column.

Q10. In one sentence each, state where you would use Oracle, PostgreSQL, MySQL and Redis at ICICI Bank, and why.

Level 2 - Applying SQL

Run each of these against `icicibank`. Include your output.

Q11. Write a query that returns the database name, the current user and the server port.

Q12. Using `information_schema`, list every column of the `loan` table with its data type and whether it accepts NULL. How many columns does `loan` have? What is the degree of the relation?

Q13. (Core) Report the number of rows in all nine tables in a single result set, sorted by table name.

Q14. List every foreign key in the database, showing the child table, the child column, and the parent table it points to. Which table has a foreign key pointing at itself, and what does that represent in the bank?

Q15. Find the three largest tables on disk, including indexes, with sizes in a human-readable form. Why is `account_txn` the largest?

Q16. (Core) ICICI Bank opens a current account for a new corporate customer. Insert the customer and the account using these details, then read both rows back.

Field	Value
customer_id	1032
name	Latha Krishnan
date of birth	1985-09-23
email	latha.krishnan@example.com
phone	9845099887
city	Chennai
kyc_status	VERIFIED
risk_rating	MEDIUM
customer_since	2026-03-01
home branch	5
account_id	9047
account_number	ICIC0000009047
account_type	CURRENT
opening balance	500000.00
opened_date	2026-03-01
interest_rate	0.00

Q17. (Core) The customer deposits 75,000 and then withdraws 20,000. Write both updates so that they compute from the current balance rather than assigning an absolute figure, then confirm the final balance is 555,000.

Q18. (Core) Try to delete customer 1032 without deleting the account first. Record the exact error. Then delete both rows in the correct order and confirm the table counts are back to 30 and 45.

Q19. Write three statements that the database must reject, one for each of these reasons, and record the error message for each:

1. An account whose `account_type` is `FIXED`.
2. A transaction whose `amount` is `-5000`.
3. A customer whose `kyc_status` is `IN_PROGRESS`.

Q20. Using two sessions, demonstrate that an uncommitted `UPDATE` in Session A is invisible to Session B. Show the SQL from both sessions and the output at each step.

Q21. (Core) In a single transaction, transfer 12,000 from account 9004 to account 9010, then roll it back. Prove with a query before and after that neither balance changed.

Q22. Show the current values of `shared_buffers`, `work_mem`, `wal_level` and `fsync`. For each, write one sentence explaining what it controls.

Level 3 - Analysis and design decisions

Answer in prose. Two to five sentences each unless stated otherwise.

Q23. ICICI Bank is launching a UPI payments product expected to peak at 4,000 transactions per second. The vendor proposes storing transactions in MySQL using the MyISAM storage engine "because it is faster for reads". Write the response you would send, naming the specific risk and the specific alternative.

Q24. The bank has 40 million rows in `account_txn` and growing by 2 million a month. The branch operations team wants a screen showing every transaction for an account, and the risk team wants monthly totals across all accounts. Explain why one database design cannot serve both well, and describe the two-system arrangement you would propose. Name which module of this programme builds each part.

Q25. (Core) During an incident review, a developer says: "The application logged 'transfer successful', so the money must have moved." Explain why that statement is not evidence, and state what would be.

Q26. ICICI Bank's DBA proposes setting `fsync = off` on the core banking PostgreSQL server because it makes overnight batch loads three times faster. Write a short recommendation. If you would allow it anywhere, say exactly where and under what conditions.

Q27. (Core) A new joiner argues that check constraints such as `balance >= 0` belong in the application, not the database, because "the application is where the business rules live". Give two arguments for keeping the rule in the database, and one situation where the application layer genuinely is the right place.

Q28. ICICI Bank is acquiring a smaller bank whose core system runs on PostgreSQL, while ICICI runs Oracle. Using only what you learned today, list four differences between the two products that you would expect to cause problems during the merge. You will test your list against reality in Module 11.

Q29. (Written, 10 to 15 lines.) Reply to ticket ICICI-4417 from the branch operations head. Explain, in language a non-technical banker will understand, what moving from spreadsheets to a relational database will fix, what it will not fix, and what the bank must do differently as a result. Do not use the words "ACID", "normalization" or "referential integrity" - explain the ideas instead.